



Short communication

Halogen-free boron based electrolyte solution for rechargeable magnesium batteries

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HIGHLIGHTS

- A new halogen-free electrolyte was synthesized for rechargeable Mg batteries.
- This electrolyte suppressed corrosion to conventional metal current collectors.
- It was confirmed that halogen in electrolyte caused corrosion of current collectors.
- Cells using the halogen-free electrolyte show reversible charge–discharge behavior.

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ABSTRACT

All halogen containing electrolytes for Mg battery are apt to corrode conventional metal current collectors. In this paper, a new type of halogen-free boron based electrolyte ($\text{Mg}[\text{Mes}_3\text{BPh}]_2/\text{THF}$) is designed and prepared. Electrochemical tests show that this electrolyte system possesses high ion conductivity ($1.5 \times 10^{-3} \text{ S cm}^{-1}$) and good Mg deposition–dissolution reversibility. More importantly, the same electrochemical window (2.6 V vs. Mg RE) of the electrolyte on Pt and stainless steel electrodes indicates that halogen-free electrolyte indeed lessens the corrosion to conventional metal current collectors. The surface morphologies of stainless steel, aluminum and copper are further observed after their anodic potentiostatic polarization in $0.25 \text{ mol L}^{-1} \text{ Mg}[\text{Mes}_3\text{BPh}]_2/\text{THF}$ electrolyte solution for 2 days. A comparison with halogen containing electrolytes proves that the presence of halogen in electrolyte is the reason for corrosion. This work provides a stepping stone for developing new halogen-free electrolyte systems for rechargeable Mg batteries.

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1. Introduction

The demand for green secondary batteries, which possess high energy density, high safety and low cost, is continuously increasing because of fossil energy shortage and environmental issues in 21st century [1–3]. Mg is an attractive anode material for its high theoretical specific capacity (2205 mAh g^{-1} , 3832 mAh cm^{-3}), and very negative standard electrode potential (-2.37 V vs. SHE) [4]. In addition, compared with lithium metal, Mg is much cheaper (1/25 of the lithium price), more easily processed, safer [5] and not plagued by dendrite formation [6], etc. Hence, rechargeable Mg

battery using Mg anode has acquired more and more attention as one of next-generation energy storage devices [7]. Despite these attractive advantages, there are several bottlenecks impeding the development of rechargeable Mg battery [8]. With respect to electrolytes, owing to the chemical activity of Mg, the passivating films formed on Mg surface cannot conduct Mg^{2+} in most polar aprotic electrolytes [9]. Therefore, the reversibly electrochemical Mg deposition in many electrolyte solutions is impossible.

Reversible Mg deposition/dissolution processes were first reported in Grignard reagent electrolytes [10,11]. However, they cannot be used for batteries because of their poor anti-oxidation capability. In 1990, Gregory et al. investigated various magnesium salts and compounds for rechargeable Mg battery electrolytes [4]. In particular, the Mg deposition/dissolution reversibility and the solubility and ionic conductivity of organoborates were examined and a Mg/Co₃O₄ cell using $\text{Mg}[\text{B}(\text{Bu}_2\text{Ph}_2)]_2$ electrolyte was cycled. Authors conjectured that the low open-circuit cell voltage could be

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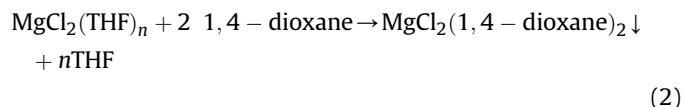
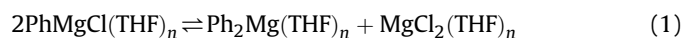
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caused by insufficient oxidative stability of the electrolyte. Aurbach et al. have proposed two kinds of Al based electrolyte solutions of $\text{Mg}(\text{AlCl}_2\text{BuEt})_2/\text{THF}$ [12] and $\text{AlCl}_3-(\text{PhMgCl})_2/\text{THF}$ [13] with high Mg deposition/dissolution reversibility and improved electrochemical windows (respectively 2.4 V and 3.3 V vs. Mg RE), which greatly promote the development of rechargeable Mg batteries. In 2011, Kim et al. [14] reported the new Al based electrolyte of $[\text{Mg}_2\text{Cl}_3-6\text{THF}][\text{HDMSAlCl}_3]$ crystal formed from the reaction of hexamethyldisilazide (HMDSMgCl) and AlCl_3 at 3:1 ratio in THF, which also possesses over 3 V window on Pt electrode and reversible Mg deposition–dissolution characteristics. Recently, our group has developed a boron based electrolyte system formed through the reaction of tri(3,5-dimethylphenyl)borane (Mes_3B) and PhMgCl in THF solvent [15,16]. It presents high ionic conductivity, excellent Mg deposition reversibility and anodic stability on Pt (3.5 V vs. Mg RE). However, it is found to be corrosive to conventional metal current collectors such as Al, Cu, Ni, stainless steel (SS) etc. One possible reason for the corrosive character is the presence of halogen in cations or/and anions of the Mg salts [17]. In order to confirm the effect of dissociated halogen in electrolytes on corrosion of conventional metals and develop anti-corrosive electrolyte systems for rechargeable Mg battery, we synthesized a new halogen-free boron based electrolyte system and investigated its electrochemical performance systematically in this paper.

2. Experimental

Mes_3B was synthesized using a modified organometallic route involving the direct reaction of organic halide, Mg, and $\text{BF}_3\cdot\text{OEt}_2$ according to reference [16]. The Ph_2Mg compound was obtained by exploiting the Schlenk equilibrium using 1,4-dioxane solvent, which forms an essentially insoluble, polymeric magnesium halide dioxane complex, $\text{MgCl}_2(1,4\text{-dioxane})_2$ (Eqs. (1) and (2)) [18]:



The halogen-free organic electrolyte of $\text{Mg}[\text{Mes}_3\text{BPh}]_2/\text{THF}$ was synthesized according to the following steps: First, Mes_3B (0.326 g 1 mmol) was dissolved in 1 mL highly pure THF solvent (water content less than 30 ppm), and then, the $\text{Ph}_2\text{Mg}/\text{THF}$ solution

(1 mol L^{-1} 0.5 mL, 0.5 mmol) was slowly added to the $\text{Mes}_3\text{B}/\text{THF}$ solution. The resulting solution was stirred for 5–10 h at room temperature. Its structural information was obtained by running multinuclear NMR spectra.

All the electrochemical measurements of the electrolyte and XRD, SEM measurements were carried out according to the previously reported literature procedure [19]. The standard CR2016 coin cells, using thin Mg disc anode, Entek PE separator, halogen-free electrolyte and Mo_6S_8 composite cathode (SS as current collector), were assembled in glove box containing less than 5 ppm water and O_2 . Mo_6S_8 Chevrel phase material was synthesized according to the reported method [20]. The charge–discharge measurements of the coin cells were carried out on a land battery measurement system (Wuhan, China) with the cutoff voltage of 1.8/0.5 V vs. Mg RE.

3. Results and discussion

3.1. Structural information

We performed a multinuclear NMR analysis for elucidating the structural information of the species in $\text{Mg}[\text{Mes}_3\text{BPh}]_2/\text{THF}$ electrolyte, which can be obtained by running spectra for each of the nuclei: ^{11}B and ^{13}C . ^{11}B NMR spectra of $\text{Mes}_3\text{B}/\text{THF}$ and $\text{Mg}[\text{Mes}_3\text{BPh}]_2/\text{THF}$ in Fig. 1 show their respective characteristic peaks at -20.2 ppm and -26.2 ppm. The later is typically ascribed to a tetra-coordinated boron anion $[\text{Mes}_3\text{BPh}]^-$ formed [16]. Here the bond of the phenyl moiety to boron was confirmed. A comparison between ^{13}C NMR spectrum (a) and (c) in Fig. 2 indicates that there was no un-reacted Ph_2Mg in the $\text{Mg}[\text{Mes}_3\text{BPh}]_2/\text{THF}$ solution. These results confirm that the $\text{Mg}[\text{Mes}_3\text{BPh}]_2$ compound was synthesized successfully via the Lewis acid–base reaction between Mes_3B and Ph_2Mg .

3.2. Mechanism of reactions

Based on the above NMR spectra, it is possible to analyze the reactions and equilibria that take place in the solution. Eqs. (3)–(6) show the proposed key steps in the formation of $\text{Mg}[\text{Mes}_3\text{BPh}]_2$. In Eqs. (3) and (4), Ph^- anion was generated by the ionization of Ph_2Mg , and the depletion of Ph^- reacted with Mes_3B in Eq. (5)

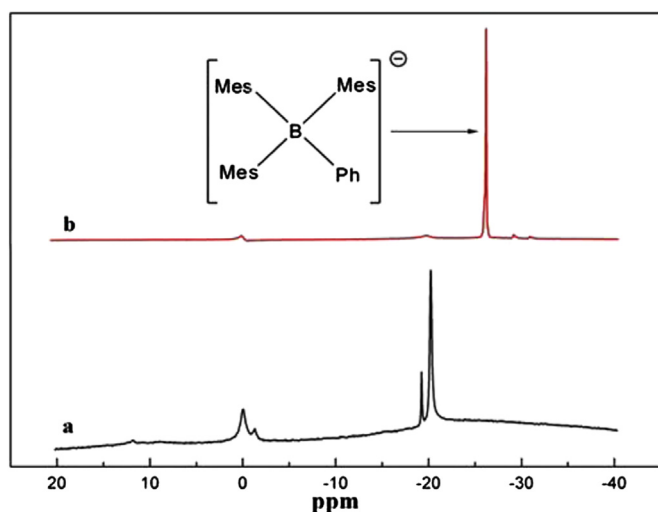


Fig. 1. ^{11}B NMR spectra measured with the following solutions: (a) 0.5 mol L^{-1} $\text{Mes}_3\text{B}/\text{THF}$, (b) 0.25 mol L^{-1} $\text{Mg}[\text{Mes}_3\text{BPh}]_2/\text{THF}$.

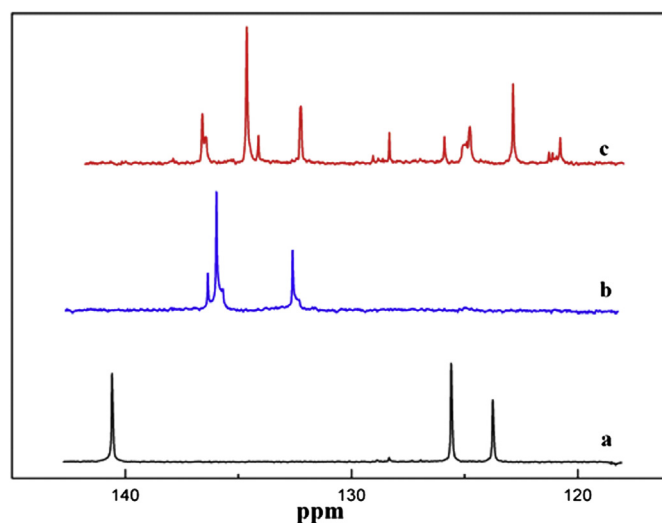


Fig. 2. Aromatic region of ^{13}C NMR spectra measured with the following solutions: (a) 0.5 mol L^{-1} $\text{Ph}_2\text{Mg}/\text{THF}$, (b) 0.5 mol L^{-1} $\text{Mes}_3\text{B}/\text{THF}$, (c) 0.25 mol L^{-1} $\text{Mg}[\text{Mes}_3\text{BPh}]_2/\text{THF}$.

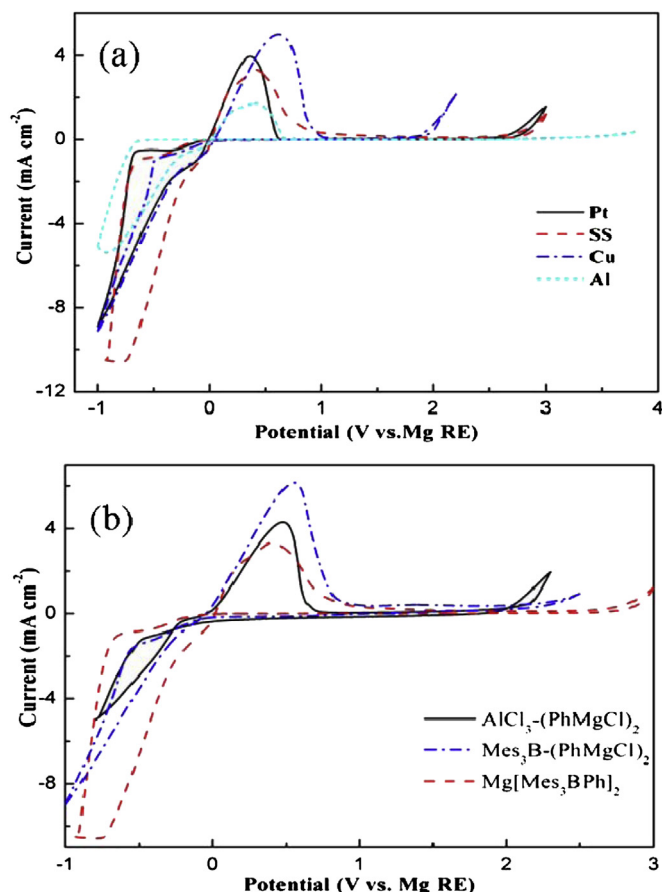
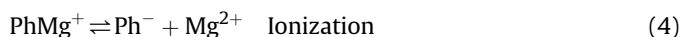
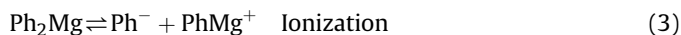


Fig. 3. (a) Typical cyclic voltammograms on different metal current collectors in the 0.25 mol L⁻¹ Mg[Mes₃BPh]₂/THF electrolyte solution, Mg ribbon as RE and CE. (b) Typical cyclic voltammograms on SS electrode in three electrolyte solutions. (Black) 0.4 mol L⁻¹ AlCl₃-(PhMgCl)₂/THF, (blue) 0.5 mol L⁻¹ Mes₃B-(PhMgCl)₂/THF and (red) 0.25 mol L⁻¹ Mg[Mes₃BPh]₂/THF. Scanning rate is 50 mV s⁻¹. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

drives the equilibrium to the right. On the basis of the three equations, a general Eq. (6) of the reaction between Ph₂Mg and Mes₃B is obtained.



3.3. Electrochemical behaviors

Ionic conductivity of 0.25 mol L⁻¹ Mg[Mes₃BPh]₂/THF solution was determined to be about 1.5×10^{-3} S cm⁻¹ at room temperature, which can meet the basic requirement of battery use. The Mg deposition reversibility and the anodic stability of Mg[Mes₃BPh]₂/THF electrolyte were evaluated by cyclic voltammetry (CV) with different working electrodes. As shown in Fig. 3(a), current peaks for Mg deposition–dissolution processes can be clearly observed with Pt, Cu, Al and SS electrodes, which indicate that halogen is not a key element to control the reversibility of Mg deposition. The

electrochemical window of the electrolyte on SS and Ni (not shown here) is about 2.6 V vs. Mg RE, almost same as that on inert metal Pt electrode, whereas the anodic stability on Al reaches to 3 V, much higher than that in halogen-containing electrolytes (for example, about 1.2 V in the Mg(AlCl₂EtBu)₂/THF [21] and (PhMgCl)₂-AlCl₃/THF [22] electrolytes). This evidence hints that the halogen-free electrolyte indeed suppresses the corrosion to conventional metal current collectors. The onset potential of 2.6 V is associated with the electrolyte decomposition. On the other hand, the anodic stability of Mg[Mes₃BPh]₂/THF electrolyte on Cu current collector is still about 1.8 V, similar to that in halogen-containing electrolytes. This phenomenon may be determined by the nature of Cu itself, not caused by electrolyte corrosion [23].

Fig. 3(b) compares typical steady-state cyclic voltammograms of SS electrode in three electrolyte systems. The electrochemical oxidative stabilities of SS in AlCl₃-(PhMgCl)₂/THF electrolyte and Mes₃B-(PhMgCl)₂/THF boron based electrolyte are 2.1 V and 2.2 V respectively, much lower than those on Pt electrodes [16]. Muldoon et al. [17] studied the corrosion of SS in 0.2 mol L⁻¹ (Mg₂Cl₃·6THF) (HMDSAICl₃)/THF electrolyte solution, and they found that corrosion pits emerged on SS working electrode at potential above 2.2 V (vs. Mg RE). Therefore, corrosion of SS by halogen-containing electrolytes limits the charging cut-off voltage of a SS coin cell to below 2.2 V. In contrast, the halogen-free Mg[Mes₃BPh]₂/THF electrolyte can extend the anodic stability window of SS to 2.6 V, which makes the conventional SS cell component useful. So far, the discharge voltage of all the reported cathodes for rechargeable Mg batteries is below 1.7 V. The anodic stability of 2.6 V is competent for these battery systems.

In general, the anodic stability of electrolytes is mainly dominated by anionic structures and interaction between anions and cations. We could speculate that the electrochemical window of Mg[Mes₃BPh]₂/THF solution was determined by tetra-coordinated boron anion [Mes₃BPh]⁻, where the complex Lewis acid may delocalize part of charge from Ph⁻ and make the [Mes₃BPh]⁻ anodically more stable than Ph⁻ (or Ph₂Mg). In fact, as shown in Fig. 4, the electrochemical window of Ph₂Mg/THF was only 1.8 V, much narrower than that of Mg[Mes₃BPh]₂/THF.

The critical parameter of coulombic efficiency in the Mg deposition/dissolution processes was obtained by galvanostatic cycling experiments, which was calculated according to the ratio of the charge amount of Mg dissolution to that of Mg deposition. As shown in Fig. 5(a), the cycle efficiency is greatly improved with the increase of cycling number. The first coulombic efficiency is about

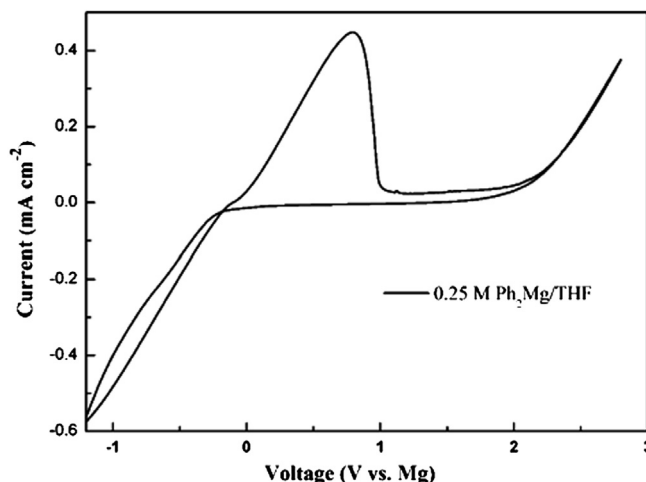


Fig. 4. Typical cyclic voltammogram on Pt in 0.25 mol L⁻¹ Ph₂Mg/THF, Mg ribbon as RE and CE. Scanning rate is 50 mV s⁻¹.

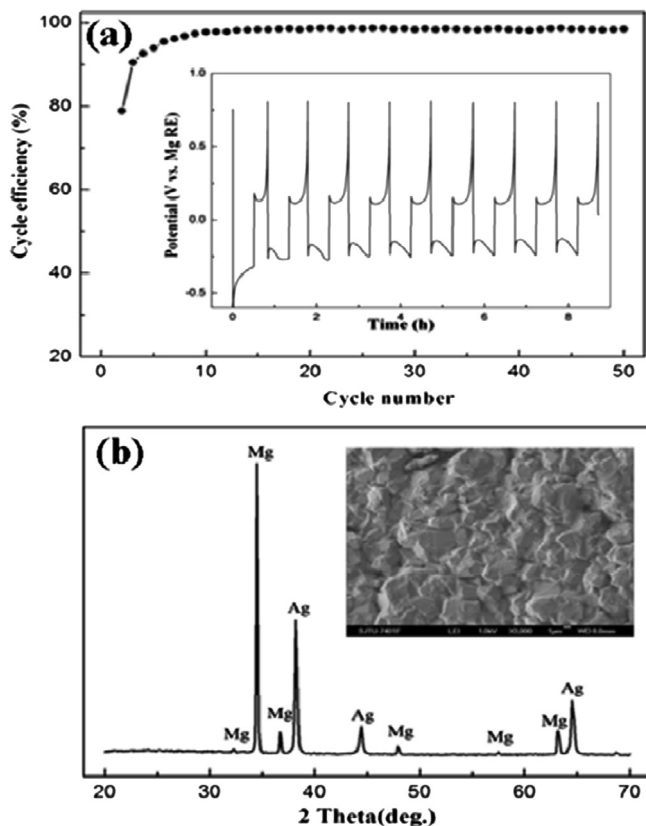


Fig. 5. (a) The cycling efficiencies and the initial nine chronopotentiograms of Mg deposition and dissolution on Ag in 0.25 mol L⁻¹ Mg[Mes₃BPh]₂/THF electrolyte solution; (b) XRD pattern of the electrodeposited Mg on a Ag substrate, and the charge amount is 3.6 C cm⁻².

80%. After 15 cycles, it approaches to 100%. The inset of Fig. 5(a) shows the initial nine chronopotentiograms of Mg deposition–dissolution in 0.25 mol L⁻¹ Mg[Mes₃BPh]₂/THF. It can be clearly observed that the overpotential for electrochemical Mg deposition–dissolution decreases gradually along with the cycle number.

The deposit layer from Mg[Mes₃BPh]₂/THF electrolyte solution was characterized by XRD and SEM measurements, as shown in Fig. 5(b). The peaks at 32.2, 34.4, 36.6, 47.8, 57.4, 63.1, 67.3 and 68.6 (2θ) in XRD pattern are in accordance with hexagonal close-packed structure of Mg (JCPDS, 350821) and the peaks at 38.2°, 44.4° and 64.6° are ascribed to Ag substrate (JCPDS, 04-0783). No peaks related to other phases can be detected, thus indicating high Mg purity of the deposit. SEM image (Fig. 5(b) inset) demonstrates the uniform morphology of Mg deposition, which is very important for the practical application of a secondary battery due to avoiding metallic dendrite formation.

To further examine the usability of the halogen-free electrolyte for a full cell with a magnesian/de-magnesian cathode, 2016-type coin cell using typical Mo₆S₈ cathode and Mg disc anode was assembled, and the charge and discharge properties were investigated at room temperature. Fig. 6(a) presents its typical potential–time curve with the discharge and charge voltage limits of 0.5/1.8 V. The cell demonstrates reversible cycling behavior at 0.05 C rate. Fig. 6(b) shows the typical capacity–voltage profile of the cell with distinct charging platform (1.3 V) and discharge platform (0.9 V). The discharge capacity is about 96 mAh g⁻¹ on the basis of the weight of Mo₆S₈. This result confirms that the halogen-free electrolyte has good compatibility with the Mo₆S₈ cathode and could be practically used in rechargeable Mg battery.

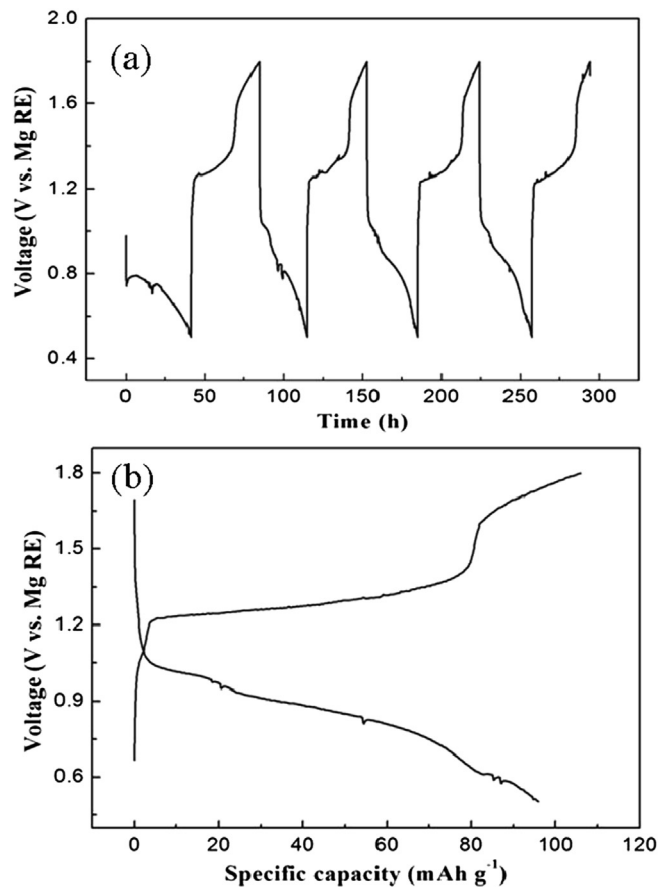


Fig. 6. Charge–discharge profiles of Mg/Mo₆S₈ coin-cells at constant current rate (0.05 C), the electrolyte is 0.25 mol L⁻¹ Mg[Mes₃BPh]₂/THF solution. Cathode current collector is SS.

3.4. Surface morphologies of metal current collectors

In order to check the anodic corrosion behavior of different metal current collectors in electrolyte, potentiostatic polarization of different metal current collectors in Mg[Mes₃BPh]₂/THF electrolyte solution was conducted. Fig. 7 compares the surface morphologies of SS, Al and Cu current collectors before and after anodic polarization. As shown in Fig. 7(a'), (b'), the surface morphologies of SS and Al change little after polarization at 2.5 V and 2.6 V for two days. In contrast, the surface of Cu was corroded seriously after polarization at 2 V for two days and many caverns can be observed (Fig. 7(c')). Furthermore, it was noticed that there was a red Cu deposit layer on the opposite electrode after opening the coin cell. This result is indicative of the electrochemical oxidation and dissolution of Cu at 2 V vs. Mg.

4. Conclusions

A halogen-free organic Mg[Mes₃BPh]₂/THF electrolyte system has been successfully synthesized using a reaction between Mes₃B Lewis acid and Ph₂Mg Lewis base. It presents high electrochemical reversibility for Mg deposition/dissolution, no corrosive nature to conventional metal current collectors, relatively high ionic conductivity (1.5 × 10⁻³ S cm⁻¹) and competent anodic stability (2.6 V vs. Mg RE). The electrochemical results reveal that the dissociated halogen in the electrolytes leads to corrosion to conventional current collectors such as SS and Al. SEM observation confirms that SS and Al in Mg[Mes₃BPh]₂/THF electrolyte are stable under potential polarization at 2.5 V and 2.6 V, respectively, while Cu could not be

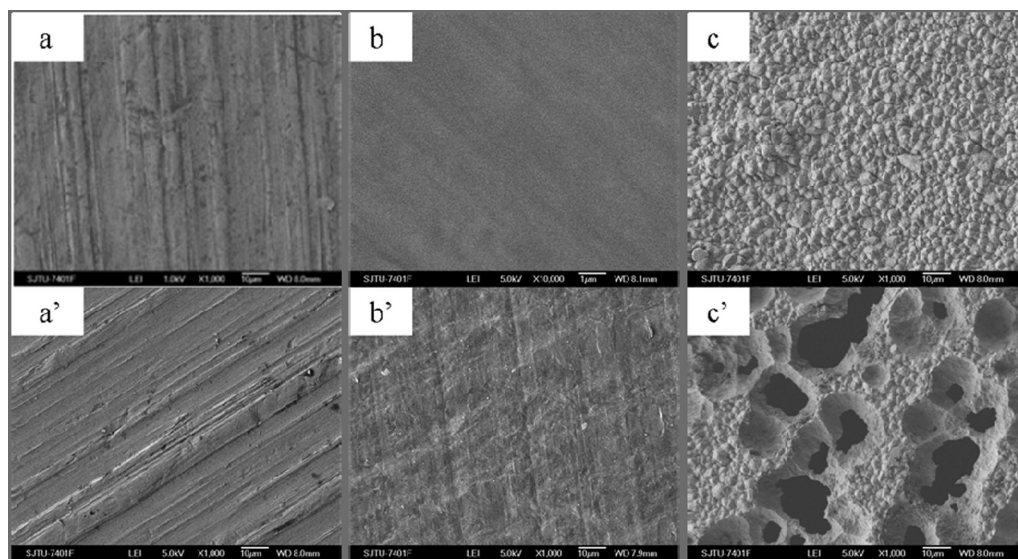


Fig. 7. SEM images of different metal current collectors before and after potentiostatic polarization in 0.25 mol L⁻¹ Mg[Mes₃BPh]₂/THF electrolyte solution for 2 days. (a) pristine SS, (a') SS, polarization potential is 2.5 V; (b) pristine Al, (b') Al, polarization potential is 2.6 V; (c) pristine Cu, (c') Cu, polarization potential is 2 V.

used as current collector for the battery with charge cut-off voltage over 2 V. Finally, the reversible charge–discharge cycling performance of Mg/Mo₆S₈ cell using SS as cathode current collector indicates that the halogen-free electrolyte could be practically used in rechargeable Mg battery.

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